

Classification unit description

Pile Driving or Dredging Classification Unit: 722004	Sector: Construction
	Sub-sector: Heavy Construction

Introduction to classification units

We group all firms into classification units based on the products produced, the services provided, and the processes, technology, or materials used. The base premium rate reflects the cost of compensation and prevention in the classification unit.

This classification unit description provides an overview of the classification unit for general guidance only. Firms are classified based on this, our classification policies, Rate and Classification List, and other relevant classification unit descriptions. The items listed below are examples only; they are intended to be illustrative (not exhaustive). Every example may not apply to every firm in the classification unit.

Description This classification unit covers firms that are engaged in ground improvement processes to ready a surface for further construction which may involve dynamic compaction, pile driving, vibroflotation, vibro-compaction, vibro-displacement, or vibro-replacement.

Also included are firms that dredge harbours and various waterways to remove bottom sediments, and those that construct containment dykes by dredging.

Build or renovate Here are some examples of buildings or structures that firms in this classification unit build or renovate:

Caissons

Footings

Services Here are some examples of services that firms in this classification unit provide:

Bore foundations
Clean waterways
Drive pier columns
Dynamic compaction
Soil densification
Vibro-compaction
Vibro-replacement
Widen harbours

Clean harbours
Creosote application
Drive sheetpile
Shore & tieback work
Underpinning
Vibro-displacement
Vibroflotation densification
Widen waterways

Equipment Here are some examples of equipment that firms in this classification unit use:

Concrete forms
Drill rig
Front-end loader
Pile driving mechanical hammer
Suction dredge

Cranes
Excavator
Pile driving gravity hammer
Pile driving vibratory hammer
Vibroflot

Occupations Here are some examples of key workers in this classification unit:

Bridgeman
Deckhand
Oiler

Crane operator
Dredge operator
Surveyor

General information

Pile Driving - drive timber or steel piles into the ground by means of striking the top of the pile or by vibration or extracting piles by pulling on the pile with a crane or by a combination of vibration and pulling with a crane.

Dynamic compaction - involves the systematic dropping of heavy weights in a pattern designed to remedy poor soil conditions at a building site.

Vibroflotation - encompasses a variety of techniques designed to densify, reinforce, and/or stabilize loose and soft soils with the insertion of vibrating poker-type equipment. Vibroflotation includes techniques such as vibro-compaction, vibro-displacement and vibro-replacement.

Vibro-compaction - a vibrating probe is inserted into the ground by both jetting and vibration. After the probe reaches the required depth of compaction, granular material is added from the ground surface to fill the void space created by the vibrator. A compacted radial zone of granular material is created.

Vibro-displacement - a process using compressed air to displace the soil laterally as a probe is advanced through the weak strata. Backfill is placed in to the hole in stages as the probe is incrementally withdrawn and lowered again to compact the fill. This process, also known as the 'dry method', forms a stone column. The columns are typically smaller in diameter than the 'wet' method and are used in the stiffer soils.

Vibro-replacement - a process of constructing stone columns using a vibratory probe (vibroflot). After the vibroflot penetrates to the desired depth of treatment, stone fill is deposited into the hole from the ground surface or through feed tubes to the tip of the vibroflot as it is withdrawn.

Hazard rating assignment

Based on the nature and extent of the risks and hazards in this classification unit, firms in this classification unit are assigned a high hazard rating for the purpose of defining first aid-related requirements (i.e., the application of G3.16, First aid assessment, in the Occupational Health and Safety Regulation). This section of the Regulation specifies mandatory minimum levels of first aid, including what type of first aid kits, facilities, and equipment are required.

When to contact us

Firms in this classification unit may also occasionally engage in other business activities. However, if they have a significant presence in any of the following industries, it may make sense for them to be assigned to another classification unit.

- 1) tunnelling (see CU 722009, CU 722010, & CU 722011); or
- 2) land clearing, site surface preparation, excavation, or related shoring work by processes other than pile driving (see CU 721031).

Please contact us if you have any questions about this or anything else related to this classification unit description.

Base premium rates

2026: 0.71% (or \$0.71 per \$100) of assessable payroll
2025: 1.09% (or \$1.09 per \$100) of assessable payroll
2024: 1.36% (or \$1.36 per \$100) of assessable payroll